Ankit Bhattacharjee

Indian Institute of Technology Kharagpur

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GitHub | Google Scholar | LinkedIn

Research Interests: Machine Learning, Natural Language Processing, Scientific Machine Learning, Fluid Mechanics.

Education

Program	Institution	%/CGPA	Year
B.Tech (Hons.), Mechanical Engineering	Indian Institute of Technology Kharagpur	8.7/10	2027

Publications & Preprints

- **A Bhattacharjee**. "Fluid Flow as Transport of Probability: Entropy, Compressibility and Irreversibility". *Physica A: Statistical Mechanics and its Applications 699 (2026) 131908*. PHYSA 131908.
- **A Bhattacharjee**, K Bhaumik. "Robust Dual-Signal Fusion: Hybrid Neuro-Symbolic Gating with Compressed Chain-of-Thought Refinement for Irony Detection in Social Media Texts". *Proceedings of the 11th Workshop on Natural User-Generated Text (W-NUT), at EMNLP 2026*. **Accepted**. arXiv preprint arXiv:2606.16845.
- **A Bhattacharjee**. "Quantifying the Occult: A Comparative Study of Hindu and Buddhist Deities Using Machine Learning Methods". *Digital Scholarship in the Humanities*. **In peer review**. arXiv preprint arXiv:2609.27074.
- **A Bhattacharjee**, S Maity, S Chakraborty, I Mallick. "Dose-PlanNet: Physics Based Radiotherapy Dose Prediction with Deep Learning". *arXiv preprint arXiv:2609.26901*.
- **A Bhattacharjee**, I Hari, K Bhaumik, A Das, P Sarangi. "Beyond Model Bias: Cross-Entropy Based Rankings in Literary and Academic Texts Remain Stable across Small Language Models". **In ACL Rolling Review (ARR), October (2026) Cycle**. *arXiv preprint arXiv:2610.17040*.

Professional Experience

Reviewer & Program Committee Member | W-NUT, EMNLP 2026 *OpenReview | June - July '26*

- Invited by the Programs Chairs of the *11th Workshop on Natural User-generated Text* to serve as a reviewer.
- Invited to serve as a reviewer for the workshop, in the next edition, to be held alongside a *CL conference in 2027.

Internships

Research Intern | Automated Radiation Treatment Planning *Tata Medical Center | May – July '26*

- Developed **'Dose-PlanNet'**, a physics-guided 3D U-Net architecture for prostate radiotherapy dose prediction.
- Deployed DL to optimize radiation dosimetry, achieving as-is clinically acceptable plans across multiple trials.

Ongoing Research Work

Stability Analysis of the Rainy–Bénard Model | BTech Project *Prof. Rajaram Lakkaraju*

- Derived the saturated drizzle equilibrium and verified its consistency with the classical Rayleigh–Bénard case.
- Analyzed linear and energy stability, and verified the supercritical transition through direct numerical simulations.

Course Work

- **Mathematics & Computing:** Machine Learning | Mathematical Statistics | Linear Algebra, Numerical & Complex Analysis | Programming & Data Structures* | Operations Research | Advanced Calculus | Theory of Partial Differential Equations | Transform Calculus | Boundary Integral Methods | Soft Computing | Data Driven Methods in Thermal & Fluid Sciences | Finite & Boundnary Element Methods in Manufacturing
- **Theoretical Physics, Fluid & Thermal Sciences:** Dynamics | Fluid Mechanics* | Basic Thermodynamics | Heat Transfer* | Applied Thermodynamics | Physics of Waves | Electrodynamics-I | Applied Elasticity

*Courses with a Laboratory component

Skills

- **Programming Languages:** Python, C++, C [*Specialist* on **Codeforces** (Handle: *SrabonGitikar*)]
- **Technical Tools & Libraries:** NumPy | Pandas | Scikit-learn | SciPy | PyTorch | PyDicom | GitHub | LaTeX
- **Agentic AI Tools:** Antigravity | Windsurf | Devin

Academic Achievements

- Secured **IITKGP's Conference Grant** to attend **EMNLP** and to present at W-NUT, collocated with EMNLP.
- Secured All India Rank of 7486 (**99.36%ile**) in JEE Main 2023 and All India Rank of 9787 in JEE Advanced 2023.
- Secured All India Rank of **199** in Indian Institute of Science Education and Research (**IISER**) Aptitude Test 2023.

Extra-Curricular Achievements

- Represented the institute in multiple FIDE rated and unrated tournaments, rated 1600+ in chess.com Rapid.
- Bronze medalist in General Championships of Cricket and Band (Music), representing MMM Hall, IIT Kharagpur.